

HX-400 Operating and Maintenance Manual

Controlled technical manual

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1. Scope and safety

This manual applies to the HX-400 High-Temperature Heavy-Duty Pump in standard Helios configurations.

Installation, commissioning and maintenance must be performed by qualified personnel. Isolate electrical and process energy before service.

2. Operating envelope

Parameter	Limit / guidance
Flow range	30-180 m³/h
Head	up to 128 m
Maximum system pressure	20 bar
Pump-body continuous temperature	140°C continuous fluid temperature
Suspended solids	Maximum suspended solids size 3 mm

3. Seal configuration

Above 130°C, PTFE seal configuration is required.

Standard configuration: FKM mechanical seal. Standard continuous-temperature guidance: 130°C continuous.

Available alternatives: PTFE up to 160°C; double mechanical seal and flush plan options.

Seal life is influenced by fluid chemistry, shaft speed, pressure, temperature cycling and dry-running events.

4. Installation

- Install on a rigid, level foundation.
- Align coupling after piping loads are connected.
- Avoid unsupported pipework loads at suction and discharge nozzles.
- Provide adequate suction conditions and avoid operation outside the recommended flow region.
- Confirm direction of rotation before introducing process fluid.

5. Commissioning checklist

Check	Acceptance criterion
Process isolation removed	Confirmed safe to open
Pump primed	Casing full; no dry running
Seal flush / support	Configured if required
Coupling alignment	Within installation tolerance
Temperature and pressure	Within documented limits

6. Maintenance

Routine inspection every 6,000 operating hours; every 3,000 hours for severe thermal cycling

Record operating temperature, pressure, vibration observations and seal leakage during planned inspections.

7. Troubleshooting

Symptom	Likely causes	First checks
Low flow	Incorrect rotation; restriction; air ingress	Check rotation, valves, suction line
High vibration	Misalignment; cavitation; pipe strain	Check alignment, NPSH conditions, supports
Seal leakage	Seal wear; incompatible material; dry running	Review service conditions and seal selection
Overheating	Operation outside envelope; poor cooling	Check duty point and fluid temperature